



Designs of zwitterionic polymers

Yung Chang¹

Received: 17 October 2021 / Accepted: 20 April 2022 / Published online: 22 June 2022
© The Polymer Society, Taipei 2022

Abstract

Zwitterionic polymers have become a kind of potential biomaterial system used to design nonfouling interfaces between general materials and biological environments. This antifouling polymer system is inspired by biology, especially due to the presence of zwitterions or alternately charged groups on cell membranes and protein surfaces. Previous systematic studies have shown that the zwitterionic polymer system can exhibit excellent anti-adsorption, anti-coagulation, anti-adhesion and anti-attachment functions for biological components such as human proteins, blood cells, tissue cells, general bacteria, and marine shellfish. Research in recent years has pointed out that the latest development of zwitterionic polymers is of great significance in the fields of bio-inert materials, dental care, biomedical devices, membrane bio-separation, medical stents, and drug delivery carriers. This review article focuses on sorting out and illustrating the research and development of molecular design, material properties, engineering process, and system applications of zwitterionic polymers, which is expected to inspire new ideas and directions in bio-inert polymer field.

Keywords Zwitterionic polymers · Nonfouling interfaces · Bio-inert materials · Antifouling coatings

Introduction

In recent years, previous studies have conducted systematic research and discussion on potential material systems that can resist the adsorption of various components in the biological environment, which has also become an important development hotspot in the field of nonfouling material engineering [1–5]. At present, three generations of antifouling material system have been evolved and developed. The first antifouling material consists of 2-hydroxyethyl methacrylate (HEMA) and its derivatives [6–8]. Abundant hydroxyl functional groups lead to form hydrogen bonds with water molecules, thereby forming a protective hydration layer that repels biofouling. However, past research tends to show that HEMA materials cannot provide the best biofouling protection in complex media, such as human blood, while HEMA-based polymer interfaces actually show quite poor biocompatibility. Therefore, HEMA materials have been gradually replaced, and further efforts have been made to

develop a material system that can exhibit better antifouling performance at interfaces regardless of the complex medium composition. Since the 1980s, the design of the second antifouling material polyethylene glycol (PEG) and polymer derivatives has become very successful in the surface PEGylation of the general material with a controlled biofouling resistance [9–13]. However, the disadvantage of PEG derivatives is that the EG group tends to decompose when in contact with oxygen-rich environment and transition metal ions, and thus the stability of the PEGylation has begun to be questioned. Since the early 2000s, the design of the third antifouling material zwitterionic betaines and polymer derivatives has become very powerful in the surface zwitterionization of the general material with excellent bio-inert interfaces [14–17]. Unlike HEMA and PEG, the stable and tight hydration layer formed on the interface modified with zwitterions comes from electrostatic interaction, leading the perfect antifouling performance of zwitterionic polymer interfaces.

In the past 30 years, zwitterionic betaine materials have been extensively researched and developed in the form of solutions, interfaces, hydrogels, and membranes systems [18–20]. So far, more than 4,500+ SCI papers about zwitterionic material research have been published in the field of chemistry, materials science, and chemical engineering. Different from the previous studies in the literature, our

✉ Yung Chang
ychang@cycu.edu.tw

¹ R&D Center for Membrane Technology and Department of Chemical Engineering, Chung Yuan Christian University, Chungli District, Taoyuan 320, Taiwan

research team has focused on the molecular design, surface modification, and membrane formation of zwitterionic sulfobetaine polymers. The biomedical applications include (1) anti-sticking dressing for wound recovery, (2) anti-bacterial coating for teeth protection, (3) anti-coagulant blood separation membrane, (4) bio-inert grafting for medical devices and (5) bio-compatible drug delivery system. The discussion of this review article firstly focuses on the design of different zwitterionic polymer formulation to summarize these works. These structural designs help to explore the outstanding antifouling properties of zwitterionic polymers. In the second part, bio-inert property of zwitterionic polymers will be introduced, including antifouling performance of biological components such as proteins, cells, blood cells, and bacteria. In the third part, we will carefully introduce the properties and types of the material structures modified by zwitterionic polymers, and the surface zwitterionization. In the fourth part, we will discuss the scope of biomedical applications that we have specifically targeted in the past 16 years [16, 17]. Finally, we will discuss the current challenges in the field of zwitterionic polymer materials, and where more efforts should be made to bring full impacts in potential application range of zwitterionic polymer, and conclude this article with a future development perspective.

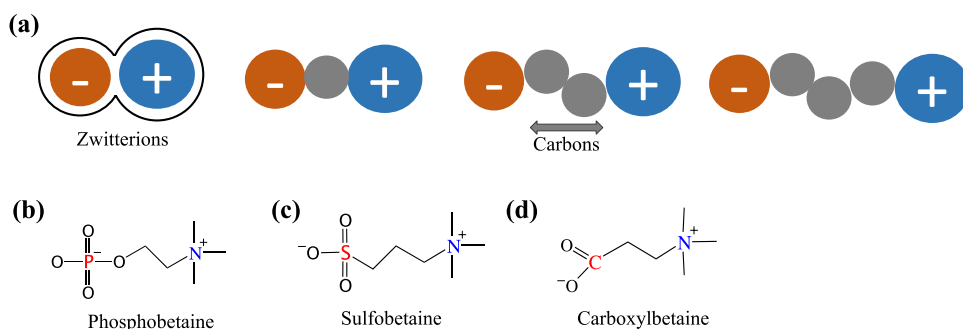
Molecular structure design of Zwitterionic polymers

From the aforementioned introduction part, it was well-known that zwitterionic polymers can produce a good hydration protective layer with water molecules, so they can become an ideal nonfouling material in a general biological environment. In recent years, zwitterionic structures have been used in bio-inert (so called anti-biofouling or biofouling resistant) polymer materials and can be applied to medical devices in contact with human blood, which has attracted great attention in the field of biomaterials engineering [21–26]. For the design of bio-inert materials, reducing protein adsorption is a key indicator for avoiding the adhesion of cell tissues and improving biocompatibility [27–30].

Past studies have pointed out that when the amount of fibrinogen adsorption is less than 10 ng/cm^2 , it can prevent the biomaterials from cell accumulating, biofilm formation and tissue fibrosis, and can also avoid the thrombus reaction [31–35]. Therefore, reducing or preventing the non-specific adsorption of proteins is of key significance for realizing the surface of bio-inert polymer materials. In general, because the bio-inert interface constructed by zwitterionic polymers is highly hydrated with water molecules through electrostatic interactions, it can form a low surface free energy state in an aqueous solution, which can effectively resist various protein adsorption, cell adhesion, and blood cell activation, as well as bacterial attachment and biofilm formation [36–45]. As shown in Fig. 1a, the major monomer structure in zwitterionic polymers is a chemical molecular unit composed of a positively charged group and a negatively charged molecular group. The distance between these two molecular groups affects the degree of hydration of the zwitterionic structure to water molecules and the corresponding biological inertness [46–50]. The chemical structures of the main zwitterions that have been widely studied, discussed and used in the literature include phosphobetaine (PB), sulfobetaine (SB) and carboxybetaine (CB), as shown in Fig. 1b–d.

The first zwitterionic monomer used to prepare zwitterionic polymers was methacryloyloxyethyl phosphorylcholine (MPC), which was a design proposed by Japanese scholar Dr. Nakabayashi in 1978 [51]. In 1990, Professor Ishihara from the University of Tokyo proposed a new concept of synthesis route and preparation method, which can break through the past reaction limits and obtain medical-grade high-purity MPC monomer (Purity > 99.9%) [52]. Therefore, many research and development work has been inspired and promoted. The various bio-inert properties of zwitterionic polymers have been discussed, and a large number of comparisons with PEG-based polymers have also been made. In 2006, Dr. Chang first published a study about the use of atom transfer radical polymerization (ATRP) to prepare the surface of sulfobetaine methacrylate (SBMA) polymer brushes, and to measure protein adsorption with a surface plasmon resonance (SPR) instrument [29]. The results found that PSBMA brushes can control the amount of fibrinogen

Fig. 1 Chemical structures of (a) zwitterions, (b) phosphobetaine, (c) sulfobetaine, and (d) carboxylbetaine



adsorption below 10 ng/cm^2 , and proved to be effective in preventing platelet adhesion and activation, and avoid blood coagulation reactions. In 2008, American scholar Dr. Jiang made a breakthrough in CBMA research [53]. In addition to proving that CBMA has good biological inertness, Jiang's group also proposed a new method for functionalization of zwitterionic structures. The carboxylic acid group in the CB structure can achieve high-efficiency molecular grafting of biomolecules, so it can expand the biological specific binding function, surpassing the single bio-inert functionality of MPC and SBMA.

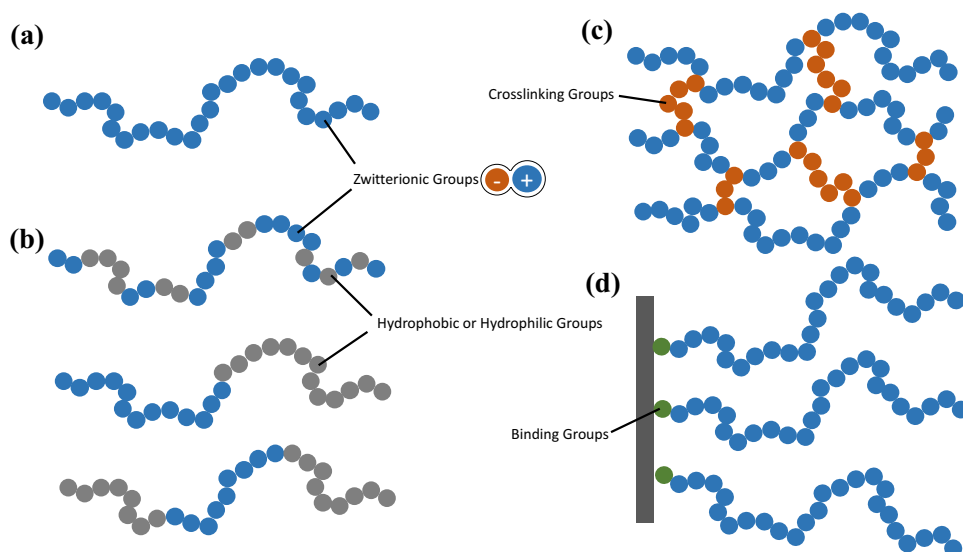
Same as general polymer material systems, zwitterionic polymers can be polymerized by monomers, and prepared into the form of homopolymers, copolymers, polymer networks and polymer brushes, as shown in Fig. 2. Past studies have reported that the use of free radical polymerization can control the molecular weight, molecular configuration, copolymer composition, crosslinking structure, and polymer brush density of zwitterionic polymers [24, 29, 33, 37, 39, 54–58].

Regarding the design, synthesis, and application of zwitterionic sulfobetaine-based polymers, we have invested in the polymerization preparation of at least 20 polymer structures in the past 16 years. As shown in Fig. 3, several important formulations can be used for medical applications. Studies have found that by adjusting the molecular weight of the zwitterionic polymer, the hydrophilicity of poly(sulfobetaine methacrylate) (PSBMA; Fig. 3a), can be effectively controlled, thereby affecting the bio-inert properties [37]. This work points out that PSBMA with a molecular weight of 135 kDa can exhibit high hydration capacity at 37°C and exert the best anticoagulant and antihemolytic activity in human blood. By introducing a hydrophobic molecular chain segment combined with PSBMA, poly(propylene oxide)-*block*-poly(sulfobetaine methacrylate) (PPO-*b*-PSBMA;

Fig. 3b) can be prepared by ATRP [29, 54]. Previous studies have found that the higher molecular weight of PPO-*b*-PSBMA copolymer-coated hydrophobic surface has higher protein adsorption, but after the surface is backfilled with the small molecular weight of PPO-*b*-PSBMA copolymer, the protein adsorption capacity can be reduced to a very low level $< 5 \text{ ng/cm}^2$. In the research on the blood compatibility of zwitterionic polymer, it has been found in the past that when the ambient temperature is too low, the contact of PSBMA with human blood will cause the plasma protein to adsorb on the polymer interface and induce coagulation. We systematically report how the morphological changes of poly(*N*-isopropyl acrylamide)-*block*-poly(sulfobetaine methacrylate) (PNIPAAm-*b*-PSBMA; Fig. 3c) copolymers in different temperature ranges affect human blood compatibility [39]. The research project shows that this nonionic-zwitterionic copolymer is a blood-compatible polymer that can potentially provide thermo-responsive behavior, and can be used for human blood purification and preservation, especially under low-temperature conditions at 4°C . When using zwitterionic polymers to form a reliable bio-inert interface, adjusting the chain length and surface packing density are two important variables. Poly(3-sulfopropyl methacrylate)-*block*-poly(sulfobetaine methacrylate) (PSA-*b*-PSBMA; Fig. 3d) was synthesized by ATRP [24]. This work constructs an ideal model platform for realizing the control of various surface packing parameters and provides a practical method for grafting zwitterionic polymer to general biomaterials through charge pairing to create a blood-inert surface in whole blood.

In the production of ceramic or metal-based medical devices, the temperature in the process can reach above 150°C . However, although zwitterionic PSBMA polymer is biologically inert as expected, it loses contamination control during high-temperature annealing. We found that the

Fig. 2 Designs of zwitterionic polymers in the form of (a) homopolymers, (b) copolymers, (c) polymer networks, and (d) polymer brushes



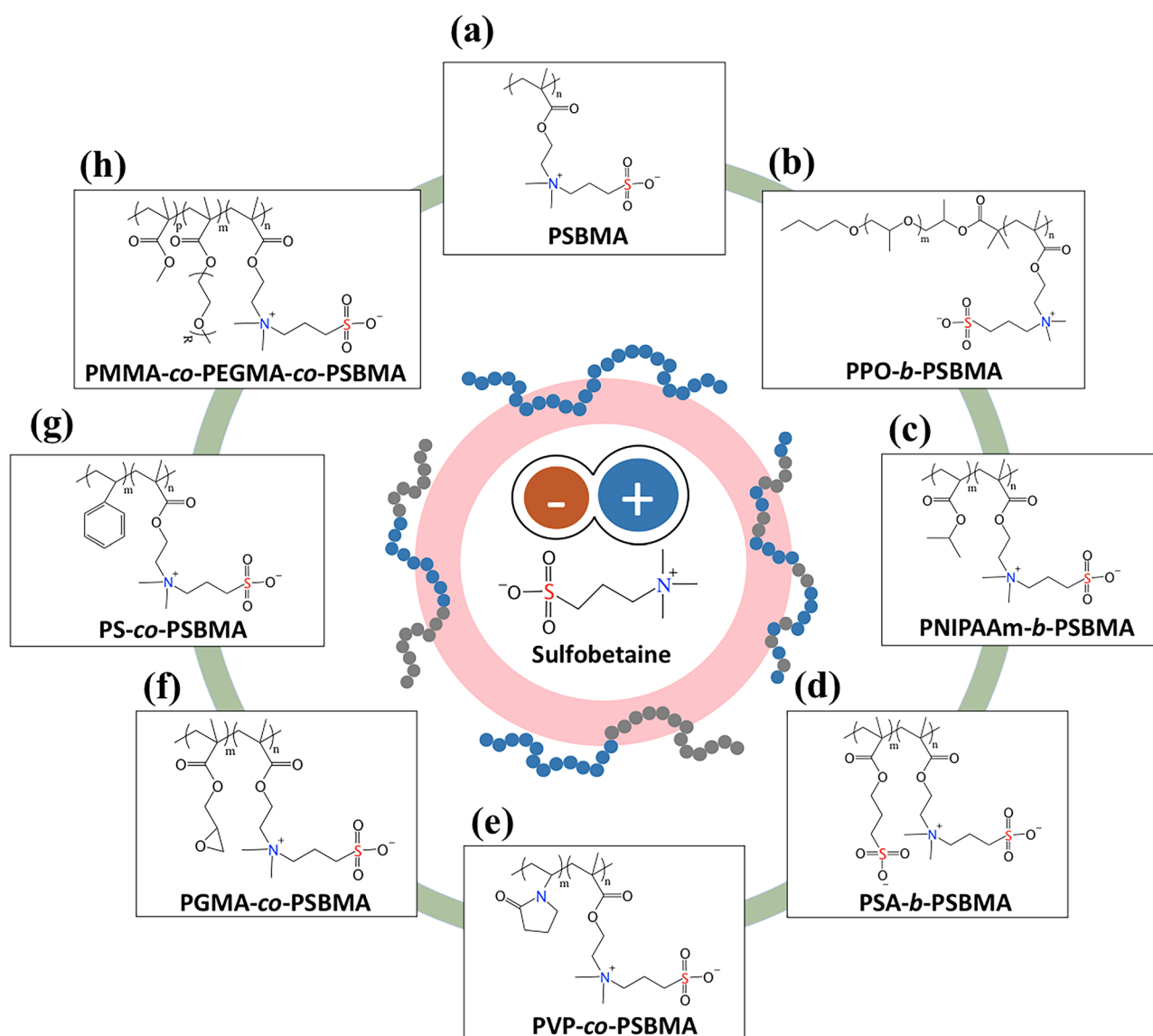


Fig. 3 Molecular structures of zwitterionic sulfobetaine-based polymers designed and studied in Dr. Chang's group

structure of the poly(vinylpyrrolidone)-co-poly(sulfobetaine methacrylate) (PVP-*co*-PSBMA; Fig. 3e) copolymer material can maintain the physical properties of the material at a high annealing temperature of 200 °C while maintaining good bio-inert performance [55]. Therefore, by combining PVP and PSBMA segments as a cross-linkable copolymer network with low deformation and good viscoelastic properties, a new type of thermosetting zwitterionic polymer material is suitable for medical material coating can be prepared. Most biomedical materials lack simple, robust and effective bio-inert modification methods in the process of surface functionalization, which limits their application potential in biomedicine. However, how to develop a general surface grafting strategy to solve this demand is a difficult challenge.

We have designed a new formula based on sulfobetaine polymer, poly(glycidyl methacrylate-co-sulfobetaine methacrylate) (PGMA-*co*-PSBMA; Fig. 3f), which can be used for surface zwitterionization of a variety of biomaterials, including ceramics, metals and plastics [33]. This type of epoxylated zwitterionic polymer provides a potential opportunity for surface grafting of zwitterions on next-generation medical devices and the development of bio-inert material interfaces.

Although sulfobetaine-based polymers have excellent anti-fouling properties, their solubility problems limit their application in surface modification of substrates by coating or blending. Our work proposes an innovative method to form a random copolymer of polystyrene (PS) and PSBMA, and

through a self-deposited modification process, the typical hydrophobic material of poly(Vinylidene fluoride) (PVDF) membrane can be modified via a combined polymerization and self-assembling coating process using polystyrene-co-poly(sulfobetaine methacrylate) (PS-*co*-PSBMA); Fig. 3g) [56]. When the monomer molar ratio of PS/SBMA is 50/50, the best bio-inert conditions can be obtained compared. According to systemic biofouling tests, the experimental results prove that it can effectively resist fibrinogen adsorption and reduce biological pollution by at least 90%, but it also eliminates blood cell adhesion (red blood cells, white blood cells, platelets, and whole blood) or bacterial attachments. PVDF polymeric membranes coated with PS-*co*-PSBMA can provide excellent antifouling performance. In addition to using a self-assembling coating process to overcome the solubility issue of zwitterionic polymers, we have also synthesized a new type of sulfobetaine-based polymer. Poly(methyl methacrylate)-co-poly(ethylene glycol methacrylate)-co-poly(sulfobetaine methacrylate) (PMMA-*co*-PEGMA-*co*-PSBMA; Fig. 3h) copolymer can be directly combined with the hydrophobic PVDF membrane material [57]. It was blended via the phase inversion process to achieve anti-biofouling bicontinuous membranes. This work proved that PVDF and PMMA-*co*-PEGMA-*co*-PSBMA zwitterionic copolymers have very good molecular-level compatibility. This is due to the use of polymer segments that are soluble with PVDF, and PEGMA can adjust the solubility of hydrophilic PSBMA segments. Through the in-situ vapor-induced phase separation (VIPS) process, bicontinuous PVDF membranes with high-flux and strong resistance to biological contamination medium can be made. These membranes have a wide range of potential applications, from protein separation to biomedical-related applications.

Blood-inert property of Zwitterionic polymers

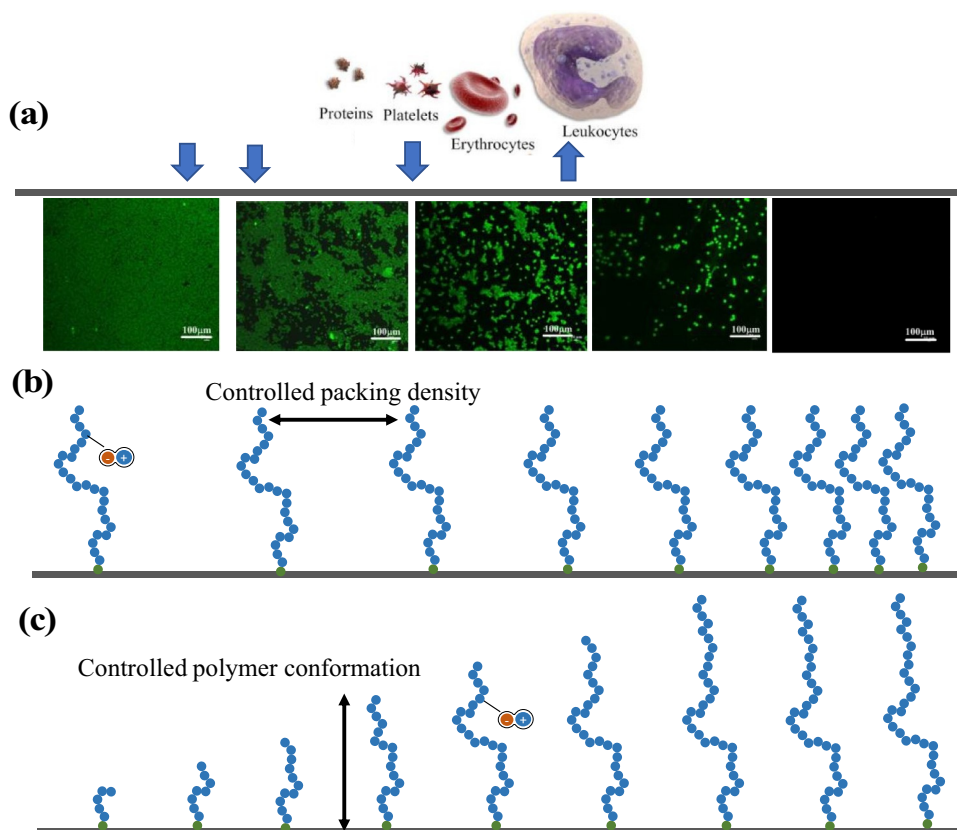
Zwitterionic polymers have attracted more and more attention as blood-inert biointerfaces and functional biomaterials because they have excellent inhibitory effects on plasma protein adsorption, platelet adhesion and activation, and thrombosis in vitro. The general feature of zwitterionic sulfobetaine-based polymers is that they have both cationic quaternary amino groups and anionic sulfate groups on the same side chain, thereby maintaining the charge neutrality of the overall polymer structure. As shown in our previous work, zwitterionic PSBMA homopolymers can generate a tightly bound structured water layer around the zwitterionic head group through electrostatically induced hydration [36, 37, 49]. In recent years, zwitterionic PSBMA polymers have been widely studied due to its ease of a scale-up synthesis. It was shown that the blood inertness of zwitterionic PSBMA

biointerfaces is attributed to their strong hydration capacity and conformational structure of water molecules.

Our previous research results pointed out that controlling the surface grafting density is very important for zwitterionic polymer surfaces with ultra-low protein adsorption properties. It was reported that when an optimized film thickness is used to control the surface grafting of the zwitterionic layer, the chemically immobilized zwitterionic brush has excellent resistance to non-specific protein adsorption from a single protein solution and complex media. As shown in Fig. 4, the packing density and polymer conformation of zwitterionic polymer brushes on the biointerface can affect platelet adhesion or human blood cell adhesion. We use ion-pair anchoring of zwitterionic brushes to systematically and accurately control chain lengths and surface packing densities of zwitterionic functional groups affect blood compatibility [24]. Our previous studies have pointed out that if the polymer chain spacing of zwitterionic brushes from biointerfaces is too large or the polymer chain length is insufficient, these factors will decrease the anti-thrombotic ability of zwitterionic interfaces in contact with human whole blood. When preparing zwitterionic brushes, it is necessary to control the optimal packing density and polymer conformation at the same time to obtain a perfect anti-blood cell adhesion biointerface, as shown in Fig. 4a. The distance between the anchored polymer chains of zwitterionic brushes must be shortened to increase the packing density of zwitterionic functional groups on the biointerface, and reduce the surface defect structures to improve the resistance of blood cell attachment, as shown in Fig. 4b. Another way is to increase the anchored polymer chain length of zwitterionic brushes to increase the coverage of zwitterionic functional groups per unit area of biointerface, as shown in Fig. 4c. Our previous studies pointed out that controlling the anchored polymer chain length of zwitterionic brushes will be easier to obtain better blood-cell resistant biointerfaces than the controlled distance between the anchored polymer chains. The result of this study is that supported by the model of PSBMA brushes directly fixed on the surface of the charged substrate through an ion pair anchoring strategy, and provides a convenient method for surface zwitterionization. We proved that when the surface is in contact with 100% human whole blood, the resulting surface fixed with zwitterionic polymer brushes maintains excellent blood inertness (“zero” blood cell adhesion).

In previous research, Jiang’s group used quantum chemistry calculations and molecular dynamic simulations to predict and explore the nano-scale effect of carbon space length on the hydration and bio-inert properties of zwitterionic molecules [41, 48]. For zwitterionic molecules with carbon space length (CSL) < 3, an increase in CSL leads to a decrease in the free energy of hydration. However, for zwitterionic molecules with CSL ≥ 3, molecular simulation

Fig. 4 Blood compatible control of zwitterionic polymer interface regulated by chemical structures, packing density, and polymer conformation. Downward arrows represent biomolecules that are adsorbed or attached to the substrate interface; upward arrows represent biomolecules that are not adsorbed or attached to the substrate interface



results cannot reveal significant changes in the free energy of hydration. The results of studies on the CB system indicate that when the CSL changes, the hydration of trimethylammonium and carboxyl groups exhibit different behaviors. The degree of hydration of the trimethylammonium group does not change with the change of CSL, but the degree of hydration of the carboxyl group increases with the increase of CSL. However, past studies have pointed out that the affinity of zwitterionic charged groups for water molecules affects the hydrophilicity of the material surface and protein adsorption. Controlling whether proteins are adsorbed on the interface of zwitterionic materials is an important indicator for regulating the bio-inert function. Therefore, when designing and synthesizing new zwitterionic structures, attention should be paid to controlling the optimized CSL. Due to the high surface free energy of most zwitterionic molecular structure interfaces in the gas phase, it is easy to adsorb unwanted substances and cause pollution, which has become a major limitation in the use of medical devices. In 2021, Chang's group proposed a new type of highly stable zwitterionic polymer structures. 4-vinylpyridine is introduced and combined with the CB molecular structure to exhibit both amphiphilic and zwitterionic characteristics [49]. For the new monomer structure 4-vinylpyridine carboxybetaine (4VPCB) with zwitterionic pyridine groups, The carbon spacing of the opposite charge groups can be

controlled by reacting iodoacetic acid, iodopropionic acid and 4-bromobutyric acid with 4-vinylpyridine. Our research results indicate that polymer brushes grafted with P4VPCB can exhibit antifouling and amphiphilic properties. The surface free energy of P4VPCB brushes can be controlled lower than that of traditional zwitterionic PSBMA brushes, so it can prevent the possibility of zwitterionic surface contamination in the air environment. Because of the carboxyl group in the molecular structure of P4VPCB, it can also be reacted and functionalized, which is helpful for biological detection with both biological inertness and specific biomolecule identification, and can further expand the applicability of zwitterionic polymers.

Bio-interface development of Zwitterionic polymers

One of the current challenges is to develop reliable blood-contacting biomaterials, using powerful precision surface modification strategies, which can effectively resist thrombosis when zwitterionic interfaces are in contact with human whole blood. In the past studies, the main modification process using zwitterionic polymers to construct functional biointerfaces can be divided into three types of strategies: coating, grafting (from or onto) and blending, as shown in

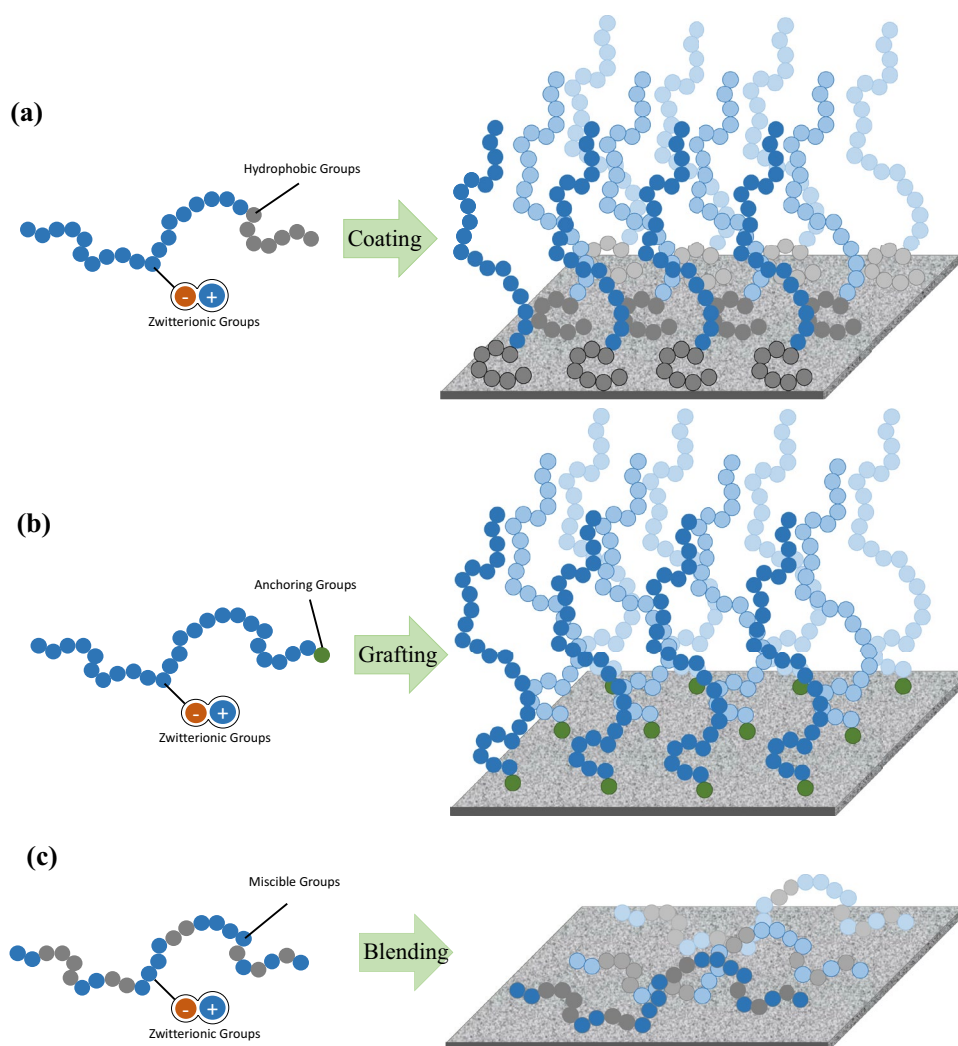
Fig. 5. In the development of these modification processes, it is necessary to consider the solubility and miscibility of zwitterionic polymers and the solvents required for the preparation process. The following content provides more detailed information about the functionalization process of these zwitterionic polymeric interfaces.

Zwitterionic polymeric interfaces functionalized by coating process

The functionalization of zwitterionic polymers on the target substrate surface by the coating process is mainly through the molecular interaction between polymer and substrate to produce physical immobilization, rather than covalent chemical bonding. Therefore, how to design suitable hydrophilic/hydrophobic segments on this type of zwitterionic polymers is quite critical. These molecular groups not only provide zwitterionic polymers to be miscible with solvents in the coating process, but also need to provide sufficient intermolecular physical binding force on the surface of the

substrate to form a stable zwitterionic coating and have good bio-inertness function, as shown in Fig. 5a. There are many methods for surface zwitterionization through well-controlled coating procedures, including dip coating, drop coating, cast coating, spray coating, deposited coating, pH-responsive coating, and phase-change coating. Among them, dip coating process is the most commonly used in the laboratory, because this procedure does not require the use of any complicated tools and is easy to be implemented. Coating density is a key parameter used to monitor the effect of surface zwitterionization. It will be greatly affected by the concentration of the copolymer in the coating solution. In the dip coating process, the target substrate is immersed in a coating bath containing an adjustable zwitterionic polymer concentration. At this stage, the density of the zwitterionic coating can be controlled according to the conditions of a given concentration of polymer solution. After completion, a thermal-induced evaporation step can be performed, in which the time for solvent removal and coating formation in the polymer solution can be controlled. The quantitative

Fig. 5 Biointerface functionalization of zwitterionic polymers from the major modification processes: (a) coating, (b) grafting, and (c) blending



control of zwitterionic coating density might be affected by the packing density and polymer conformation, and thus affect the bio-inert property of zwitterionic polymer interfaces. According to past experience, we have designed zwitterionic block copolymers and performed the dip-coating process. The well-defined zwitterionic diblock copolymers of poly(propylene oxide) (PPO) and PSBMA (PPO-*b*-PSBMA) were synthesized by ATRP polymerization method and varying two block lengths of PPO or PSBMA and ratios of PPO/PSBMA [29]. In this study, by adjusting the concentration of the polymer solution of PPO-*b*-PSBMA from 0.01 mg/mL to 1 mg/mL, the corresponding coating density will be between 0.02 mg/cm² and 0.18 mg/cm². For porous substrates, because the coating solution partially penetrates into the pores, it is difficult to accurately estimate the coating density. Generally, the surface area of the continuous pores inside the porous substrate needs to be considered. According to previous reports, for surface zwitterionization of porous membranes, when the coating density > 0.2 mg/cm², excellent biological inertness control can be obtained, which indicates that a lower coating density value can be expected to be effective for dense substrates [59].

Since most of the general substrates have hydrophobic properties, in the design of zwitterionic polymers that can be coated, a hydrophobic functional group, such as a long carbon chain or a benzene ring structure, will be introduced into the hydrophobic groups of polymer chains. If a zwitterionic block copolymer is used, a dipping solution with an appropriate polymer concentration can generate a hydrophobic force to drive the target substrate to produce a single-layer adsorption of zwitterionic polymers. The estimation of the coating density of the well-defined block copolymer will conform to the theoretical calculation. However, for the zwitterionic random copolymers, the hydrophobic interaction between the polymer segments may be entangled with the configuration of the polymer chains to form a multilayer coating, which complicates the estimation of the coating density. When using zwitterionic polymers to form a bio-inert interface layer on the surface of various substrates, the most challenging problem is how to ensure the stability of the coating without falling off due to environmental influences, which is also a major weakness of the coating process. To overcome the stability of the zwitterionic coating, it is necessary to carefully consider the strength of the molecular interaction between the modified surface and the hydrophobic unit in the copolymer structure while the proper design of the zwitterionic polymer structure. At the same time, it is also necessary to consider that the polar interaction between the zwitterionic unit facing the water phase and the surrounding medium will affect the stability of the polymer coating. In our previous studies, it has been proposed that after the coating process of zwitterionic polymers, the cross-linking reaction between polymer chains can be used

to enhance the overall stability of the polymer layer coating. Except for hydrophobic substrates that can be used for surface zwitterionization through a coating process, most natural biomedical materials have hydrophilic properties. Therefore, how to perform the coating process of surface zwitterionization on a hydrophilic substrate is a challenge for coating technology. Our study has tried to synthesize PSA-*b*-PSBMA as shown in Fig. 3d to modify nature biomedical materials with positively charged hydrophilic properties, such as chitosan or hydroxyapatite. The self-assembled coating is generated by the electric charge driven by the zwitterionic polymer containing the negatively charged block of PSA-*b*-PSBMA and thus the coated zwitterionic layer is formed. Its stability is ensured by the electrostatic interaction between the negatively charged units of PSA-*b*-PSBMA and the positively charged units of the hydrophilic substrate. In general, using a coating method to apply a zwitterionic coating layer to the target substrate can bring advantages such as simple procedure, low cost, short time, and easily scale-up process. Thus, coating process is the most powerful strategy to perform surface zwitterionization for biointerfaces [24].

Zwitterionic polymeric interfaces functionalized by grafting process

Unlike using physical coating to produce zwitterionic biointerfaces on the target substrate, the grafting process involves the formation of chemical covalent bonds between the interfaces to be modified with zwitterionic polymers. These chemical bonds provide strong molecular bonding energy and are expected to lead a high stability for preventing the zwitterionic grafting layer from leaching or falling off during operation. Grafting process can be divided into two types: (1) *Grafting from*- this method is to generate reactive sites on the surface of the target substrate first, and then use surface-initiated free radical polymerization or plasma-induced polymerization to obtain zwitterionic grafting layer. The grafting density of zwitterionic groups on target surfaces can be controlled by the monomer concentration, polymerization temperature, reaction time, and solvent in the reaction solution, which will be related to the biological inertness mentioned earlier. We have successfully tested the surface modification of SPR sensor chip, titanium surface, or porous PVDF membrane by this *grafting from* process and successfully obtained zwitterionic grafting layer. In the adsorption test of various biological components, it can perfectly demonstrate its bio-inert interfaces, especially for contacting with human blood. (2) *Grafting onto*- this method is to first synthesize zwitterionic polymers containing reactive group that can combine with the surface of the target substrate, and then anchoring zwitterionic layers, as shown in Fig. 5b. The grafting coverage is generally lower than the *grafting*

from method. Therefore, it is necessary to consider the well-controlled polymerization of zwitterionic polymers to have a suitable molecular weight range to perform excellent bio-inert, not only depends on the chemical properties of the anchoring unit, but also depends on the control of the grafting conditions. Recently we designed a new type of zwitterionic polymer to realize the surface zwitterionization process of *grafting onto*. We successfully prepared zwitterionic copolymers of poly(glycidyl methacrylate-co-sulfobetaine methacrylate) (poly(GMA-co-SBMA)) and grafted onto various target surfaces, including glass, silicon, silver, steel, titanium or polystyrene [33]. These target substrates are first activated by ozone/ultraviolet treatment. Then the ring-opening reaction catalyzed by trimethylamine occurs when the poly(GMA-co-SBMA) copolymer is grafted onto the substrates [60–64]. We use similar methods to graft GMA-co-SBMA triblock copolymers onto various metals and exhibit good bio-inert interfaces [64]. Therefore, the grafting technology can better control the chain length of the copolymer and better understand its chain configuration, because the physical properties of zwitterionic polymers can be fully characterized before the grafting step. This technique requires the use of anchor blocks that can react with surface functional groups for modification. As we know, surface zwitterionization improves its bio-inert performance by forming a hydrated layer. Therefore, most of the exposed surfaces to be modified are hydrophobic materials without containing commonly used functional groups, such as hydroxyl or amine groups, which are used to react with the anchor segments of the designed zwitterionic copolymers. Therefore, the material must be activated to generate these functional groups. The activated material is then coated with designed zwitterionic copolymers, and covalent bonds are formed under specific pH or temperature conditions. The grafting coverage of zwitterionic layers in well-controlled grafting process can show excellent blood compatibility.

Zwitterionic polymeric interfaces functionalized by blending process

In addition to applying coating or grafting processes for surface zwitterionization, we also studied the blending process of preparing membranes by phase inversion of zwitterionic polymer solution to control bio-inert property on the interfaces of target substrates. This procedure is to blend the membrane polymer, zwitterionic polymer, and solvent, and then expose the casting solution to a non-solvent to form a bio-inert membrane interfaces in one step. This method can not only modify the surface of the membrane, but also modify the pores of the membrane. However, since zwitterionic polymer is a highly polar material and will not dissolve in the common solvent of membrane polymers, it is necessary to add miscible segments to zwitterionic polymer to control its dissolution properties.

In recent years, our research team has proposed two molecular design strategies. The first strategy is to conduct blending research work with PVDF membrane materials for zwitterionic phosphobetaine polymers. We collaborated with Professor Ishihara to use a phosphobetaine-based copolymer containing carbamate groups, namely poly(2-methacryloyloxyethyl phosphorylcholine-co-ethylacryloyloxyethyl butylurethane) (PMBU) [65, 66]. These groups provide zwitterionic polymers with flexibility in solubility and help them to be miscible with PVDF in N-Methyl-2-Pyrrolidone (NMP). Through the vapor-induced phase inversion (VIPS) process, zwitterionic PVDF membranes blended with PMBU can be formed. In terms of anti-fouling performance, this type of zwitterionic porous membrane can exhibit outstanding fouling resistance when performing biofouling tests with whole blood, human plasma or bacteria solution. The second strategy is to conduct blending research work with PVDF membrane materials for zwitterionic sulfobetaine polymers. By adding miscible segments of PEGMA to SBMA-based polymer and changing the molar ratio of SBMA/PEGMA, we found a composition range that can be dissolved in N-methyl-2-pyrrolidone and dimethylformamide together with PVDF materials. In the mixture, it can undergo VIPS process to form zwitterionic PVDF membranes blended with PS-r-PEGMA-r-PSBMA or PMMA-r-PEGMA-r-PSBMA. In a series of bio-inert tests, this type of porous membrane can also exhibit excellent bio-inert functions.

Biomedical applications of Zwitterionic polymers

In the past 15 years, we have conducted systematic research and analysis on the chemical reactions, physical properties, biocompatibility, blood inertness, cytotoxicity and antibacterial properties of zwitterionic polymers and their derived biological interfaces. We have further expanded many practical applications and solved different types of anti-biofouling problems in medical clinics. Although the fouling in the biological medium in which the material interface is immersed may have many different sources, no matter whether it is a biomaterial designed for *in vivo* or *in vitro* related applications, the mechanisms required to resist biofouling are very similar. Therefore, the development of a zwitterionic interface with universal bio-inert control on different-type substrates is a key step in the biomedical applications of zwitterionic polymeric materials, and it is also a major challenge.

The detection of the protein adsorption on the substrate surface is an important indicator for evaluating the general antifouling performance of the zwitterionic interface. This is because proteins are responsible for the biofouling components at the nanoscale and are directly related to large-scale biofouling caused by organisms (such as bacteria or cells) at a macroscale. In the past studies, after surface

modification of the model substrates by zwitterionic polymers, we first measured the adsorption status from various types of proteins, the typical methods including ultraviolet optical density measurement, enzyme-linked immunosorbent assay (ELISA), and surface plasma resonance analysis, etc. [67–74]. Blood cells and tissue cells are the first major category of biological contaminants, and the measurement of the fibrinogen adsorption on the material surface can evaluate its anti-blood cell adhesion. In addition, bacteria are the second largest category of bio-foulants, and it is necessary to avoid bio-film formation on biomaterial interfaces for medical-related applications. The measurement of the adsorption of albumin on the biomaterial surfaces can provide a reliable evaluation index. Next, we will sort out the results of using zwitterionic polymers in biomedical applications,

including (1) anti-sticking dressing for wound recovery, (2) anti-bacterial coating for teeth protection, (3) anti-coagulant blood separation membrane, (4) bio-inert grafting for medical devices and (5) bio-compatible drug delivery system, as shown in Fig. 6.

Anti-sticking dressing for wound recovery

The skin is the body's first line of defense against the invasion of pathogens outside the body, and it is also the soft tissue system with the largest surface area and high elasticity in human organs. When the skin is damaged due to external force impact, tissue wounds of different degrees of damage will be produced, and subsequent incomplete repair or complete regeneration of subsequent healing will occur due to

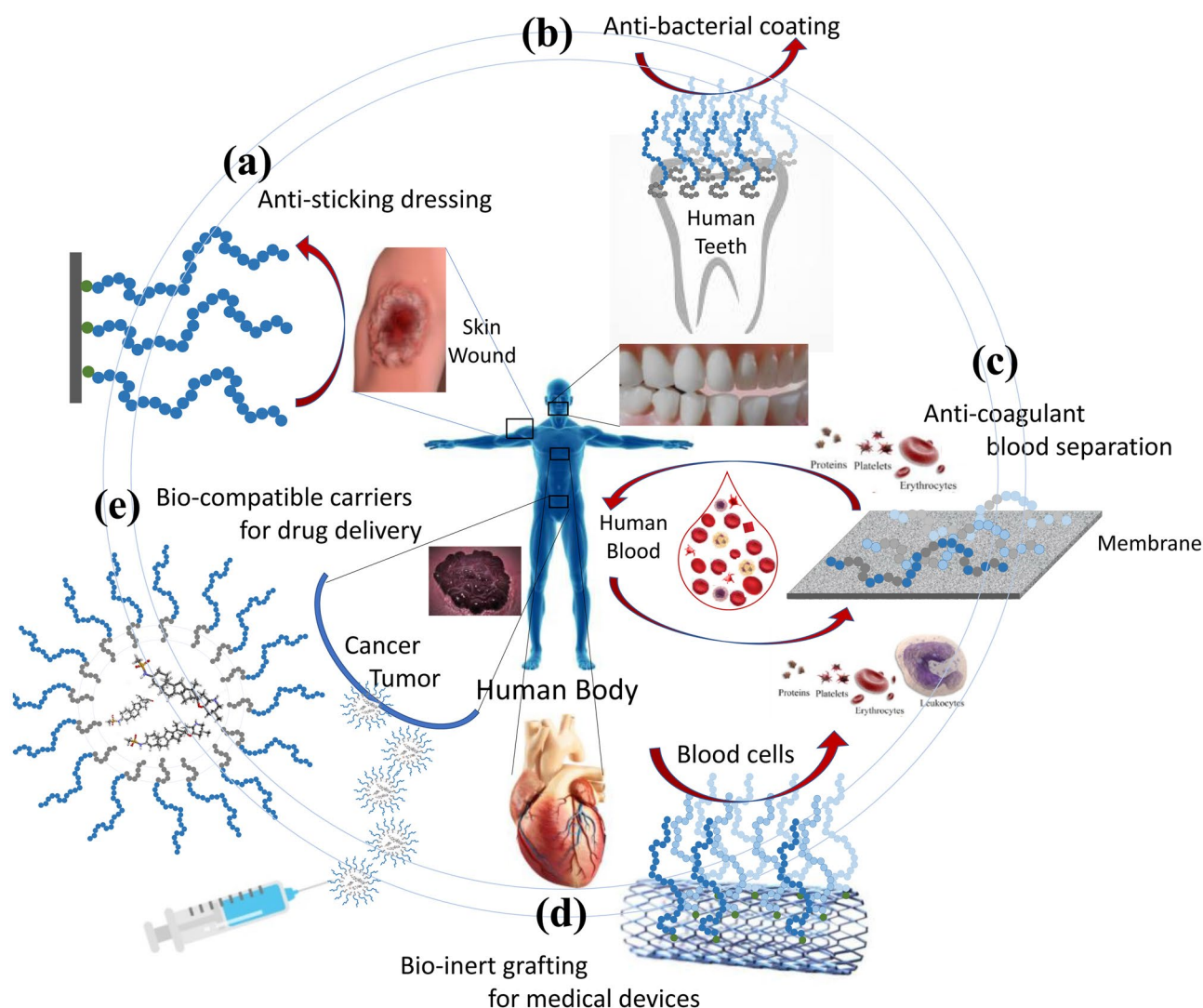


Fig. 6 Biomedical applications of Dr. Chang's zwitterionic polymers: (a) anti-sticking dressing for wound recovery, (b) anti-bacterial coating for teeth protection, (c) anti-coagulant blood separation membrane,

(d) bio-inert grafting for medical devices and (e) bio-compatible drug delivery system

the difference in structure and tissue morphology. After skin tissue ruptures in some cases, biodegradable scaffolds can be used to induce the growth of damaged tissue cells, so that the wound is healed closer to complete regeneration. At present, the best biodegradable scaffold system is the wound dressing system containing extracellular matrix (ECM). When the damaged area of the wound is too severe, such as the destruction of the dermis or hair follicle, the cells in the tissue lose their ability to grow and regenerate themselves. At this point, you can try to use advanced polymer dressings to assist in the regeneration of skin wounds. Based on past experience, we propose that zwitterionic polymer can design a new type of wound dressing that mimics the structure of a biomembrane, and use animal models to verify the characteristics of accelerating wound healing and regeneration [66, 75–80]. In order to promote healing and shorten wound recovery time, the design of zwitterionic dressings needs to consider a combination of multiple properties, including moisturizing, antibacterial, anti-tissue adhesion, high air permeability, and low deformation. Our study in 2013 was the first to publish a zwitterionic expanded poly(tetrafluoroethylene) (ePTFE) membranes that can be used as a new dressing to enhance wound healing and shorten healing time [75]. This work uses atmospheric plasma-induced surface zwitterionization to prepare PSBMA-grafted ePTFE membrane. Due to high hydration capability and excellent non-bioadhesive character, zwitterionic ePTFE-g-PSBMA dressings permitted a complete recovery after 14 days for deep skin wound healing and provided a potential way for use in skin tissue regeneration [76]. Our research in 2014 found that it was positively charged [2-(methacryloyloxy)ethyl] trimethylammonium (TMA) and negatively charged 3-sulfopropyl methacrylate (SA) of the new mixed-charge PSA-co-PTMA matrices. In addition, we have also observed interesting phenomena that may be of great significance in the special case of severe bleeding from the wound. When PSA-co-PTMA dressings have a slight positive charge deviation, they can provide wounds to accelerate epithelial regeneration and enhance blood coagulation.

In recent years, we have gradually invested major research resources on the healing and control of diabetic wounds. Because diabetic wound healing is a chronic inflammation process, and the fibroblast stops dividing and new blood vessels no longer grow, the tissue of this type of wound is not easy to regenerate. When diabetic patients suffer from vascular disease and peripheral neuritis complications, it is easy to cause chronic ulcers in the wounds of the feet, and severely cause amputation of the extremities. Our research in 2019 was the first to publish on the impact of controlling the molecular zeta potential changes in the zwitterionic polymer structure on the healing of diabetic wounds and constructed a healing kinetics model to optimize different wound healing stages [77–80]. The most important result

of this study pointed out that a positive charge bias of zwitterionic dressing can shorten the inflammatory period of the wound, accelerate healing and promote angiogenesis in the early stage of wound healing. The latest research points out that zwitterionic nanofibrous membranes prepared by zwitterionic polymers through the electrospinning process have significant help and potential for accelerating the healing of diabetic wounds.

Anti-bacterial coating for teeth protection

Teeth are calcified tissues in the human mouth that are used to pronounce, chew food, and fight enemies. They are composed of complex structures with different densities and hardness. Teeth contain millions of more complex nervous systems. Damaged teeth can affect human health, diet, and disease prevention. In the past, in dental treatment, in addition to repairing damaged parts, implants or dentures were also developed to replace missing teeth. In ancient times, ivory or shells were used to replace incisors. Dental caries is a common chronic disease that causes damage to the dental system and inflammation of the tissues around the teeth. Severe conditions can lead to tooth loss and the formation of tooth abscesses. Dental caries (commonly known as tooth decay) is mainly caused by the decomposition of simple sugars in food residues on the surface of teeth by bacteria, which leads to the dissociation of tooth minerals and the reduction of saliva remineralization. Therefore, the key to solving this problem is to construct a method to prevent bacteria from attaching and eroding the tooth surface. As mentioned above, the biologically inert zwitterionic polymer provides a potentially effective antimicrobial coating for dental materials [10, 64]. Our research in 2014 published the first zwitterionic coating material of polyethyleneimine and poly(sulfobetaine methacrylate) (PEI-g-PSBMA) [10]. This zwitterionic copolymer is coated on hydroxyapatite and teeth through a charge-driven self-assembly process, which can reduce the adhesion of *Streptococcus mutans* by 70%. Recently, we have proposed a new zwitterionic triblock copolymer made from glycidyl methacrylate (GMA) and sulfobetaine methacrylate (SBMA) monomers, called PSBMA-*b*-PGMA-*b*-PSBMA. The copolymer grafted to titanium root implants has been shown to resist the adhesion of bacteria and fibroblasts [64]. This result indicates that zwitterionic polymers can be further used in the coating of tooth root implants to achieve a biologically inert interface structure and prevent infections caused by cell-metal interactions.

Anti-coagulant blood separation membrane

Human blood is a kind of connective tissue that circulates in the vascular cavity of the body. Blood cells and plasma

can send oxygen, nutrients and water to various organs, and carry the metabolic waste of cells and carbon dioxide away from the tissues. The main function of blood can provide the body to carry out the transportation of various substances, the maintenance of internal environment heat, the control of immune response, the induction of blood coagulation, and the repair of blood vessel damage. When poor biocompatible materials come into contact with human blood, plasma proteins and blood cells are biological contaminants that easily adhere to the material interfaces and membranes. In particular, the attachment and activation of platelets can cause blood to coagulate. Because zwitterionic polymer has excellent blood-inert function, it can not only repel proteins, but also repel various blood components. It can provide various membrane material interfaces to reduce plasma protein adsorption, blood cell adhesion and red blood cell rupture. In recent years, we have been committed to developing the characteristics of zwitterionic polypropylene (PP) membranes, zwitterionic PVDF membranes, and zwitterionic PTFE membranes with good blood compatibility [65, 81–88]. Among them, we synthesized poly(glycidyl methacrylate-co-sulfobetaine methacrylate) (PGMA-co-PSBMA) and coated zwitterionic copolymers on the PP membranes to completely inhibit the activation of platelets in the blood [82]. According to this function, we further use zwitterionic PP membranes for the removal of leukocytes and can be used in blood transfusion procedures. Although blood transfusion therapy can save lives, the transfusion of all blood components may cause acute or delayed adverse reactions, and most adverse transfusion reactions are related to the transfusion of allogeneic leukocytes, including non-hemolytic fever, acute lung injury after transfusion, ineffective platelet transfusion, cytomegalovirus (CMV) infection, and increased chance of bacterial infection after surgery, etc. Therefore, if the blood product is used to reduce the leukocytes before storage, the adverse reactions of blood transfusion can be greatly reduced. From the research results in 2019, we first proposed the zwitterionic PP membrane filter for whole blood filtration and proved that it can effectively remove leukocytes (> 99.99%) without changing the erythrocytes concentration in the permeate. In summary, the combination of zwitterionic polymers and membrane surface modification, this coating and cross-linking approach shows great potential for improving the blood-inert performance and blood separation of the polypropylene membrane currently used for leukodepletion.

Bio-inert grafting for medical devices

Medical devices are used for disease prevention, diagnosis, monitoring and therapy in medical work. The

above-mentioned leukocyte removal filter is included in disease prevention medical equipment, which is used for blood transfusion safety. Medical equipment used for disease diagnosis and monitoring includes diagnostic equipment, medical optical equipment, electronic equipment, medical X-ray auxiliary analysis equipment, and magnetic resonance equipment. Since the development of medical devices, the largest investment has been disease treatment tools, including various surgical equipment, medical lasers, high-energy rays, nuclide equipment, extracorporeal circulation equipment, blood processing equipment, implant materials, artificial organs, operating rooms, emergency rooms, and infusion equipment, etc. In the development of many medical devices used to sustain human life systems have always been the focus of global R&D investment. Over the years, we have continuously invested in the research and development of bio-inert stents that can be used for tissue lesions in the body, and introduced zwitterionic polymer coating technology [22, 24, 30, 33, 55, 60–64, 84, 89, 90]. A stent is an implanted tubular device used for surgical treatment of diseased tubes such as blood vessels, esophagus, or ureters in the body. The stent is implanted in the patient to restore the normal delivery function of the pipeline. Symptoms that can be treated include vascular embolism, urinary tract occlusion, and narrowing of the esophagus or trachea. The stent is made of stainless steel, cobalt-chromium alloy or nickel-titanium alloy. There have different stent structures: traditional stents (bare metal stents), coated stents, covered stents, and stent vessels. The main problem encountered at present is how to overcome blood embolism or tissue adhesion caused by long-term use of a stent in the human body. Our proposed strategy is to construct a zwitterionic stent graft to overcome this dysplastic neck. After years of hard work, we first proposed a new type of zwitterionic polymer that can be used for the surface coating of metal stents. The copolymer is polymerized with vinylpyrrolidone (VP) and SBMA monomers to control the optimized VP/SBMA composition [55]. PVP-co-PSBMA can be produced in proportion. Since the molecular structure of VP can greatly improve the heat resistance of the traditional zwitterionic molecular structure, the first zwitterionic stent can be constructed at a process temperature of 200 °C. This study confirmed that the poly(VP-co-SBMA) copolymer network can be used as a blood-compatible coating on the stainless steel surface of a metal stent, and can be used for the treatment of vascular infarction. In addition, other medical devices such as surgical instruments or detection and diagnosis also require bio-inert functional coating interfaces. Therefore, we propose another copolymer of glycidyl methacrylate (GMA) and SBMA monomers, which can produce PGMA-co-PSBMA with an optimized GMA/SBMA composition ratio [33]. We verified that this epoxylated zwitterionic copolymer provide excellent antifouling properties on various ceramic, metal and plastic substrate surfaces, and

further tested the anti-scalp or anti-tissue adhesion of dental mirrors and surgical blades.

Bio-compatible drug delivery system

In terms of designing delivery microparticle carriers with targeted drug delivery functions, ‘Liposome’ is the longest-studied drug delivery system (DDS) since the 1960s. The main dosage form structure is to encapsulate drug molecules in vesicles formed by a phospholipid double-layer membrane. In the carrier preparation, cholesterol will also be added to change the phase transition temperature of the lipid membrane and increase its stability. Liposome-based DDS has become the basis for the design of various delivery vehicles. Factors to be considered include good formulation properties, targeting properties, long-term effects, reduction of drug toxicity, and improvement of drug stability. In recent years, many research teams have invested in polymer-based DDS, which has also become one of the options for targeted carrier systems. Judging from the current research stage, this type of DDS still has certain limitations, including problems such as high immune responsive activation and circulatory metabolic side effects. Our team has invested in developing zwitterionic polymers as the shielding interface of the new generation DDS carrier system for the past ten years, which can not only reduce the immune response, but also improve the specific targeting of the carrier delivery [37, 39, 91–100]. We made a zwitterionic carrier to synthesize foreign DNA, enter the desired host cell as part of the vector, and try to deliver the gene to the genome of the target cell. Cationic polymers are ideal carrier candidates for gene delivery because they can carry negatively charged gene fragments and can produce efficient foreign gene delivery. However, the cationic carrier has a strong positive charge, which cause high cytotoxicity. We have proposed a method to reduce the toxicity of the carrier by appropriately designing the core–shell composite structure, and use zwitterionic polymers to shield the positive charge from the cell toxicity and the activation of the immune system. This core–shell composite structure has two parts: poly(ethyleneimine) (PEI) or poly(2-dimethylaminoethyl methacrylate) (PDMAEMA) as the core and zwitterionic diblock copolymer of poly(acrylic acid)-block-poly(sulfobetaine methacrylate) (PAA-*b*-PSBMA) as the shell [96]. The PAA block can provide molecular electrostatic force to bind with PEI/DNA or PDMAEMA/DNA, while the PSBMA block can provide bio-inert functions to reduce the cytotoxicity of gene carriers and enhance blood compatibility. Among them, the more innovative molecular design is that the PAA block can be unprotected in acidic solutions, and the DNA complex endowed to the core can deliver genes to target cells. Therefore, the results of this study provide a potential way for the development of high-efficiency and non-toxic pH-sensitive zwitterionic gene

carriers, and also verify that zwitterionic polymers have the opportunity to be used in the development of bio-compatible drug delivery system.

Prospective directions of Zwitterionic polymer research and development

Although several research teams that focus on zwitterionic polymer within the research scope, including us, have achieved many goals related to the bio-inert control of zwitterionic biointerfaces, more research items are still needed to improve the applicability of zwitterionic materials and the value of derivative applications [101–115]. According to the statistical analysis of the ‘Scopus’ paper database, in the past 20 years (2001–2020), nearly 3,700 SCI papers have published related studies on zwitterionic polymers. Among them, about 480 SCI papers reported fouling issue studies, as shown in Fig. 7a. The entire study of zwitterionic polymers has grown substantially since 2011. Judging from the publication trend in the field of zwitterionic materials, research works have begun to receive a great of strong attention and entered a stage of continuous growth, which also represent the potential use of zwitterionic polymers in anti-fouling or bio-inert materials that will become more and more important in the next 10 years. In this review, we summary a systematic introduction about molecular design, anti-fouling property, bio-/blood-inert characteristics, biomedical applications of sulfobetaine-based polymers. Since 2001, about 780 SCI papers have reported the research work of this zwitterionic polymer system, and more than 50% of the articles have illustrated the potential directions in clinical medicine, as shown in Fig. 7b. Our research team also made case studies on the application of these issues in wound healing, dental care, blood filtration, bio-inert medical devices, and drug delivery system.

The industrial-scale synthesis and production of zwitterionic polymers are still very difficult topics. Several issues remain to be resolved, including reaction solvents, storage of reactants, product purification, product yield, and overall cost. As shown in Fig. 8, traditional general-purpose medical products that have been widely used around the world, such as dental care appliances, medical goggles, anti-bacterial masks, medical gloves, contact stethoscopes and thermometers, medicine preservation bottles and injection syringes, scalpel, infusion bag, infusion tube, blood bag, etc. These medical products use a lot of plastic materials such as polyvinyl chloride (PVC), polyethylene (PE), polypropylene (PP), polycarbonates (PC), polydimethylsiloxane (PDMS), and polytetrafluoroethylene (PTFE) but lack biocompatibility [116–155]. In the future, if the direction of research and development can make zwitterionic polymers mass-produced through

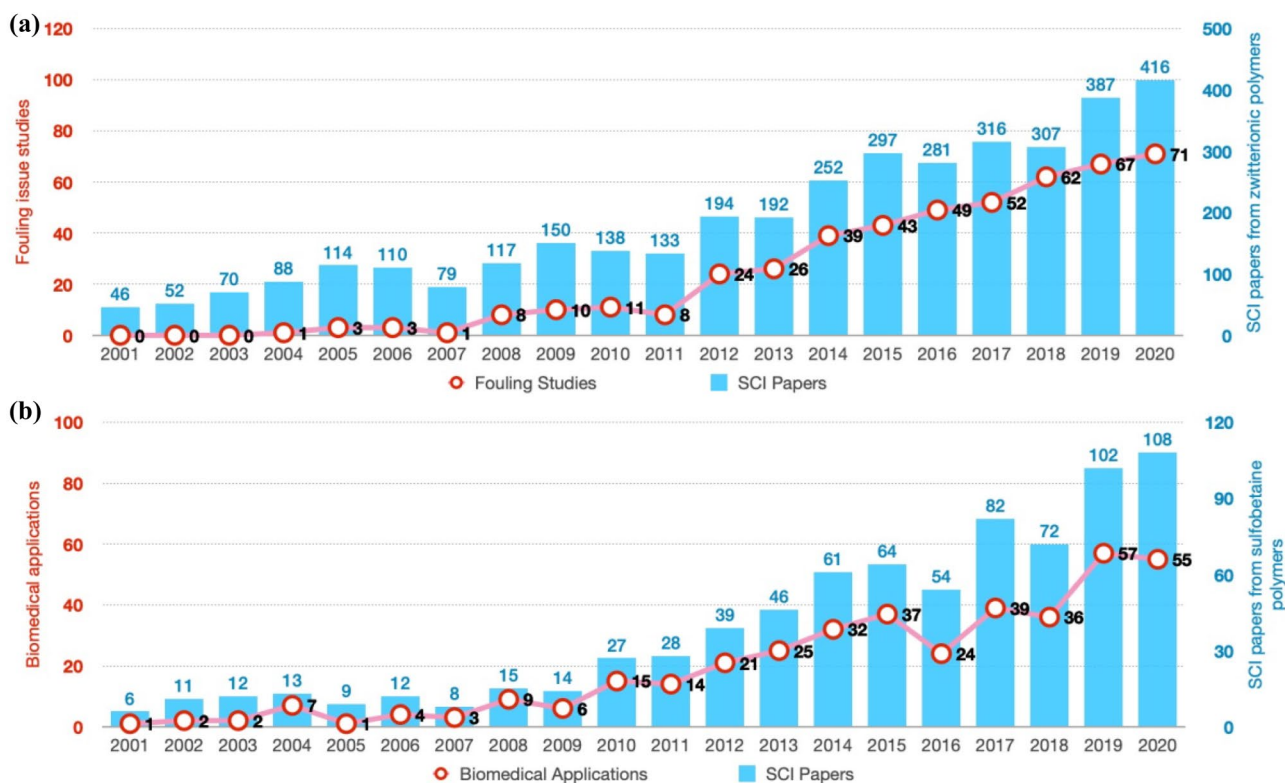


Fig. 7 The quantitative evaluation of SCI papers published in the literature from 2001 to 2020: **(a)** Research and publication about zwitterionic poly-

mers and fouling issue studies; **(b)** Research and publication about sulfobetaine polymers and biomedical applications. [according *Scopus* system]

Fig. 8 Zwitterionic polymers provide bio-inert functions and extend their practical usefulness in typical medical products made of general-purpose plastics



advanced fabrication processes, they can be functionally modified with these traditional plastics. If the bio-inert functions of zwitterionic polymers can be introduced, the safety and derivative application value of traditional general medical products can be greatly improved. In the future, the molecular structure design and synthesis process of zwitterionic polymers will be towards using green solvent and strengthening temperature resistance to meet the needs of the new generation of medical materials production and manufacturing and also to protect the environment.

Acknowledgements The author would like to thank the Ministry of Science and Technology of Taiwan.

Author contribution This manuscript was written through contributions of Dr. Yung Chang.

Declarations

Competing interest The author declare n competing financial interest.

References

- Chen S, Li LY, Zhao C, Zheng J (2010) Surface hydration: Principles and applications toward low-fouling/nonfouling biomaterials. *Polymer* 51:5283–5293
- Merian T, Goddard JM (2012) Advances in nonfouling materials: Perspectives for the food industry. *J Agric Food Chem* 60:2943–2957
- Chen SF, Zheng J, Li LY, Jiang SY (2005) Strong resistance of phosphorylcholine self-assembled monolayers to protein adsorption: Insights into nonfouling properties of zwitterionic materials. *J Am Chem Soc* 127:14473–14478
- Chen SF, Jiang SY (2008) A new avenue to nonfouling materials. *Adv Mater* 20:335–338
- Chang Y, Chu WL, Chen WY, Zheng J, Liu LY, Ruaan RC, Higuchi A (2010) A systematic SPR study of human plasma protein adsorption behavior on the controlled surface packing of self-assembled poly(ethylene oxide) triblock copolymer surfaces. *J Biomed Mater Res* 93A:400–408
- Zhao C, Li L, Yu Q, Zheng J (2011) Effect of film thickness on the antifouling performance of poly(hydroxy-functional methacrylates) grafted surfaces. *Langmuir* 27:4906–4913
- Chen H, Zhao C, Zhang M, Chen Q, Ma J, Zheng J (2016) Molecular understanding and structural-based design of polyacrylamides and polyacrylates as antifouling materials. *Langmuir* 32:3315–3330
- Montheard JP, Chatzopoulos M, Chappard D (2006) 2-Hydroxyethyl Methacrylate (HEMA): Chemical properties and applications in biomedical fields. *J Macromol Sci Part C* 23:1–34
- Chang Y, Ko CY, Shih YJ, Quemener D, Deratani A, Wei TC, Wang DM, Lai JY (2009) Surface grafting control of PEGylated poly(vinylidene fluoride) antifouling membrane via surface-initiated radical graft copolymerization. *J Membr Sci* 345:160–169
- Venault A, Yang HS, Chiang Y, Lee BS, Ruaan RC, Chang Y (2014) Bacterial resistance control on mineral surfaces of hydroxyapatite and human teeth via surface charge-driven antifouling coatings. *ACS Appl Mater Interfaces* 6:3201–3210
- Chang Y, Cheng TY, Shih YJ, Lee KR, Lai JY (2008) Biofouling-resistance expanded poly(tetrafluoroethylene) membrane with a hydrogel-like layer of surface-immobilized poly(ethylene glycol) methacrylate for human plasma protein repulsions. *J Membr Sci* 323:77–84
- Peppas NA, Hilt JZ, Khademhosseini A, Langer R (2006) Hydrogels in biology and medicine: from molecular principles to biotechnology. *Adv Mater* 18:1345–1360
- Knop K, Hoogenboom R, Fischer D, Schubert US (2010) Poly(ethylene glycol) in drug delivery: Pros and cons as well as potential alternatives. *Angewandte Chemie-International Edition* 49:6288–6308
- Mi L, Jiang SY (2014) Integrated antimicrobial and nonfouling zwitterionic Polymers. *Angewandte Chemie-International Edition* 53:1746–1754
- Shao Q, Jiang SY (2014) Molecular understanding and design of zwitterionic materials. *Adv Mater* 27:15–26
- Sin MC, Chen SH, Chang Y (2014) Hemocompatibility of zwitterionic interfaces and membranes. *Polym J* 46:436–443
- Venault A, Chang Y (2019) Designs of zwitterionic interfaces and membranes. *Langmuir* 35:1714–1726
- Lowe AB, McCormick CL (2002) Synthesis and solution properties of zwitterionic polymers. *Chem Rev* 102:4177–4189
- Laschewsky A (2014) Structures and synthesis of zwitterionic polymers. *Polymers* 6:1544–1601
- Jiang SY, Cao ZQ (2010) Ultralow-fouling, functionalizable, and hydrolyzable zwitterionic materials and their derivatives for biological applications. *Adv Mater* 22:920–932
- Chang Y, Chang WJ, Shih YJ, Wei TC, Hsiue GH (2011) Zwitterionic sulfobetaine-grafted poly(vinylidene fluoride) membrane with highly effective blood compatibility via atmospheric plasma-induced surface copolymerization. *ACS Appl Mater Interfaces* 3:1228–1237
- Chang Y, Liao SC, Higuchi A, Ruaan RC, Chu CW, Chen WY (2008) A highly stable nonbiofouling surface with well-packed grafted zwitterionic polysulfobetaine for plasma protein repulsion. *Langmuir* 24:5453–5458
- Yang W, Zhang L, Wang SL, White AD, Jiang SY (2009) Functionalizable and ultra stable nanoparticles coated with zwitterionic poly(carboxybetaine) in undiluted blood serum. *Biomaterials* 30:5617–5621
- Chang Y, Shih YJ, Lai CJ, Kung HH, Jiang SY (2013) Blood-inert surfaces via ion-pair anchoring of zwitterionic copolymer brushes in human whole blood. *Adv Func Mater* 23:1100–1110
- Vaisocherova H, Yang W, Zhang Z, Cao ZQ, Cheng G, Pilarik M, Homola J, Jiang SY (2008) Ultralow fouling and functionalizable surface chemistry based on a zwitterionic polymer enabling sensitive and specific protein detection in undiluted blood plasma. *Anal Chem* 80:7894–7901
- Ishihara K (2019) Blood-compatible surfaces with phosphorylcholine-based polymers for cardiovascular medical devices. *Langmuir* 35:1778–1787
- Inoue Y, Ishihara K (2010) Reduction of protein adsorption on well-characterized polymer brush layers with varying chemical structures. *Colloids Surf, B* 81:350–357
- Ishihara K, Nomura H, Mihara T, Kurita K, Iwasaki Y, Nakabayashi N (1998) Why do phospholipid polymers reduce protein adsorption? *J Biomed Mater Res, Part A* 39:323–330
- Chang Y, Chen SF, Zhang Z, Jiang SY (2006) Highly protein-resistant coatings from well-defined diblock copolymers containing sulfobetaines. *Langmuir* 22:2222–2226
- Chiang YC, Chang Y, Higuchi A, Chen WY, Ruaan RC (2009) Sulfobetaine-grafted poly(vinylidene fluoride) ultrafiltration membranes exhibit excellent antifouling property. *J Membr Sci* 339:151–159

31. Zhang Z, Chen SF, Chang Y, Jiang SY (2006) Surface grafted sulfobetaine polymers via atom transfer radical polymerization as superlow fouling coatings. *J Phys Chem B* 110:10799–10804
32. Chang Y, Shu SH, Shih YJ, Chu CW, Ruaan RC, Chen WY (2010) Hemocompatible mixed-charge copolymer brushes of pseudozwitterionic surfaces resistant to nonspecific plasma protein fouling. *Langmuir* 26:3522–3530
33. Chou YN, Wen TC, Chang Y (2016) Zwitterionic surface grafting of epoxylated sulfobetaine copolymers for the development of stealth biomaterial interfaces. *Acta Biomater* 40:78–91
34. Hsiao SW, Venault A, Yang HS, Chang Y (2014) Bacterial resistance of self-assembled surfaces using PPOm-b-PSBMA(n) zwitterionic copolymer - Concomitant effects of surface topography and surface chemistry on attachment of live bacteria. *Colloids Surf, B* 118:254–260
35. Chang Y, Chang Y, Higuchi A, Shih YJ, Li PT, Chen WY, Tsai EM, Hsiue GH (2012) Bioadhesive control of plasma proteins and blood cells from umbilical cord blood onto the interface grafted with zwitterionic polymer brushes. *Langmuir* 28:4309–4317
36. Wu J, Lin WF, Wang Z, Chen SF, Chang Y (2012) Investigation of the hydration of nonfouling material poly(sulfobetaine methacrylate) by low-field nuclear magnetic resonance. *Langmuir* 28:7436–7441
37. Shih YJ, Chang Y (2010) Tunable blood compatibility of poly-sulfobetaine from controllable molecular-weight dependence of zwitterionic nonfouling nature in aqueous solution. *Langmuir* 26:17286–17294
38. Chang Y, Chen WY, Yandi W, Shih YJ, Chu WL, Liu YL, Chu CW, Ruaan RC, Higuchi A (2010) Dual-thermoreponsive phase behavior of blood compatible zwitterionic copolymers containing nonionic poly(N-isopropyl acrylamide). *Biomacromol* 10:2092–2100
39. Shih YJ, Chang Y, Deratani A, Quemener D (2012) Schizophrenic hemocompatible copolymers via switchable thermoresponsive transition of nonionic/zwitterionic block self-assembly in Human Blood. *Biomacromol* 13:2849–2858
40. Shih YJ, Chang Y, Quemener D, Yang HS, Jhong JF, Ho FM, Higuchi A, Chang Y (2014) Hemocompatibility of polyampholyte copolymers with well-defined charge bias in human blood. *Langmuir* 30:6489–6496
41. He Y, Hower J, Chen SF, Bernards MT, Chang Y, Jiang SY (2008) Molecular simulation studies of protein interactions with zwitterionic phosphorylcholine self-assembled monolayers in the presence of water. *Langmuir* 24:10358–10364
42. Shao Q, He Y, White AD, Jiang SY (2010) Difference in hydration between carboxybetaine and sulfobetaine. *J Phys Chem B* 114:16625–16631
43. Leng C, Han XF, Shao Q, Zhu YH, Li YT, Jiang SY, Chen Z (2014) In situ probing of the surface hydration of zwitterionic polymer brushes: Structural and environmental effects. *J Phys Chem C* 118:15840–15845
44. Del Grosso CA, Leng C, Zhang KX, Hung HC, Jiang SY, Chen Z, Wilker JJ (2020) Surface hydration for antifouling and bioadhesion. *Chem Sci* 11:10367–10377
45. Ishihara K, Mu MW, Konno T, Inoue Y, Fukazawa K (2017) The unique hydration state of poly(2-methacryloyloxyethyl phosphorylcholine). *J Biomater Sci Polym Ed* 28:884–899
46. Huang CJ, Mi L, Jiang SY (2012) Interactions of alginate-producing and -deficient *Pseudomonas aeruginosa* with zwitterionic polymers. *Biomaterials* 33:3626–3631
47. Shao Q, Jiang SY (2014) Influence of charged groups on the properties of zwitterionic moieties: a molecular simulation study. *J Phys Chem B* 118:7630–7637
48. Li GZ, Xue H, Gao CL, Zhang FB, Jiang SY (2010) Nonfouling polyampholytes from an ion-pair comonomer with biomimetic adhesive groups. *Macromolecules* 43:14–16
49. Chiu CY, Chang Y, Liu TH, Chou YN, Yen TJ (2021) Convergent charge interval spacing of zwitterionic 4-vinylpyridine carboxybetaine structures for superior blood-inert regulation in amphiphilic phases. *J Phys Chem B*. <https://doi.org/10.1039/D1TB01374B>
50. Liu YL, Zhang D, Ren BP, Gong X, Xu LJ, Feng ZQ, Chang Y, He Y, Zheng J (2020) Molecular simulations and understanding of antifouling zwitterionic polymer brushes. *J Phys Chem B* 8:3814–3828
51. Kasoma Y, Nakabayashi N, Eiichi M, Junuchu Y (1978) Synthesis and hemolysis test of the polymer containing phosphorylcholine groups. *Kōbunshi Rombun Shū* 35:423–427
52. Ishihara K, Aragaki R, Ueda T, Watanabe A, Nakabayashi N (1990) Reduced thrombogenicity of polymers having phospholipid polar groups. *J Biomed Mater Res* 24:1069–1077
53. Zhang Z, Zhang M, Chen SF, Horbetta TA, Ratner BD, Jiang SY (2008) Blood compatibility of surfaces with superlow protein adsorption. *Biomaterials* 29:4285–4291
54. Venault A, Chang Y, Yang HS, Lin PY, Shih YJ, Higuchi A (2014) Surface self-assembled zwitterionization of poly(vinylidene fluoride) microfiltration membranes via hydrophobic-driven coating for improved blood compatibility. *J Membr Sci* 454:253–263
55. Chou YN, Chang Y, Wen TC (2014) Applying thermosettable zwitterionic copolymers as general fouling-resistant and thermal-tolerant biomaterial interfaces. *ACS Appl Mater Interfaces* 7:10096–10107
56. Venault A, Chou YN, Wang YH, Hsu CH, Chou CJ, Bouyer D, Lee KR, Chang Y (2018) A combined polymerization and self-assembling process for the fouling mitigation of PVDF membranes. *J Membr Sci* 547:134–145
57. Dizon GV, Lee YS, Venault A, Maggay IV, Chang Y (2021) Zwitterionic PMMA-r-PEGMA-r-PSBMA copolymers for the formation of anti-biofouling bicontinuous membranes by the VIPS process. *J Membr Sci* 618:118753
58. Nehache S, Yeh CC, Semsarilar M, Deratani A, Chang Y, Quemener D (2016) Anti-bioadhesive coating based on easy to make pseudozwitterionic RAFT block copolymers for blood-contacting applications. *Macromol Biosci* 16:57–62
59. Chiag YC, Chang Y, Chen WY, Ruaan RC (2012) Biofouling resistance of ultrafiltration membranes controlled by surface self-assembled coating with PEGylated copolymers. *Langmuir* 28:1399–1407
60. Tan CME, Dizon GV, Chen SH, Venault A, Chou YN, Tayo L, Chang Y (2020) Temperature-triggered attachment and detachment of general human bio-foulants on zwitterionic polydimethylsiloxane. *J Phys Chem B* 8:8853–8863
61. Fowler PMPT, Dizon GV, Tayo LL, Caparanga AR, Huang J, Zheng J, Aimar P, Chang Y (2020) Surface zwitterionization of expanded poly(tetrafluoroethylene) via dopamine-assisted consecutive immersion coating. *ACS Appl Mater Interfaces* 12:41000–41010
62. Ladd J, Zhang Z, Chen S, Hower JC, Jiang SY (2008) Zwitterionic polymers exhibiting high resistance to nonspecific protein adsorption from human serum and plasma. *Biomacromol* 9:1357–1361
63. Tang SH, Domino MY, Venault A, Lin HT, Hsieh C, Higuchi A, Chinnathambi A, Alharbi SA, Tayo LL, Chang Y (2019) Bioinert control of zwitterionic poly(ethylene terephthalate) fibrous membranes. *Langmuir* 35:1727–1739
64. Chou YN, Venault A, Cho CH, Sin MC, Yeh LC, Jhong JF, Chinnathambi A, Chang Y, Chang Y (2017) Epoxylated zwitterionic triblock copolymers grafted onto metallic surfaces for general biofouling mitigation. *Langmuir* 33:9822–9835
65. Venault A, Hsu CH, Ishihara K, Chang Y (2018) Zwitterionic bi-continuous membranes from a phosphobetaine copolymer/poly(vinylidene fluoride) blend via VIPS for biofouling mitigation. *J Membr Sci* 550:377–388

66. Maggay IV, Venault A, Fang CY, Yang CC, Hsu CH, Chou CY, Ishihara K, Chang Y (2021) Zwitterionized nanofibrous poly(vinylidene fluoride) membranes for improving the healing of diabetic wounds. *ACS Biomater Sci Eng* 7:562–576
67. Huang CJ, Wang LC, Shyue JJ, Chang YC (2014) Developing antifouling biointerfaces based on bioinspired zwitterionic dopamine through pH-modulated assembly. *Langmuir* 30:12638–12646
68. De Vera JS, Venault A, Chou YN, Tayo L, Chiang HC, Aimar P, Chang Y (2017) Self-cleaning interfaces of polydimethylsiloxane grafted with pH-responsive zwitterionic copolymers. *Langmuir* 35:1357–1368
69. Maggay IV, Suba MCAM, Aini HN, Wu CJ, Tang SH, Aquino RB, Chang Y, Venault A (2021) Thermostable antifouling zwitterionic vapor-induced phase separation membranes. *J Membr Sci* 627:119227
70. Dizon GV, Chou YN, Yeh LC, Venault A, Huang J, Chang Y (2018) Bio-inert interfaces via biomimetic anchoring of a zwitterionic copolymer on versatile substrates. *J Colloid Interface Sci* 529:77–89
71. Yang CC, Lo CT, Luo YL, Venault A, Chang Y (2021) Thermally stable bioinert zwitterionic sulfobetaine interfaces tolerated in the medical sterilization process. *ACS Biomater Sci Eng* 7:1031–1045
72. Venault A, Subarja A, Chang Y (2017) Zwitterionic polyhydroxybutyrate electrospun fibrous membranes with a compromise of bioinert control and tissue-cell growth. *Langmuir* 33:2460–2471
73. Hsu CH, Venault A, Zheng HZ, Lo CT, Yang CC, Chang Y (2021) Failure of sulfobetaine methacrylate as antifouling material for steam-sterilized membranes and a potential alternative. *J Membr Sci* 620:118929
74. Venault A, Lai MW, Jhong JF, Yeh CC, Yeh LC, Chang Y (2018) Superior bioinert capability of zwitterionic poly(4-vinylpyridine propylsulfobetaine) withstanding clinical sterilization for extended medical applications. *ACS Appl Mater Interfaces* 10:17771–17783
75. Jhong JF, Venault A, Hou CC, Chen SH, Wei TC, Zheng J, Huang J, Chang Y (2013) Surface zwitterionization of expanded poly(tetrafluoroethylene) membranes via atmospheric plasma-induced polymerization for enhanced skin wound healing. *ACS Appl Mater Interfaces* 5:6732–6742
76. Jhong JF, Venault A, Liu L, Zheng J, Chen SH, Higuchi A, Huang J, Chang Y (2014) Introducing mixed-charge copolymers as wound dressing biomaterials. *ACS Appl Mater Interfaces* 6:9858–9870
77. Venault A, Lin KH, Tang SH, Dizon GV, Hsu CH, Maggay IVB, Chang Y (2020) Zwitterionic electrospun PVDF fibrous membranes with a well-controlled hydration for diabetic wound recovery. *J Membr Sci* 598:117648
78. Lin HT, Venault A, Chang Y (2019) Zwitterionized chitosan based soft membranes for diabetic wound healing. *J Membr Sci* 591:117319
79. Venault A, Bai YW, Dizon GV, Chou HYE, Chiang HC, Lo CT, Zheng J, Aimar P, Chang Y (2019) Healing kinetics of diabetic wounds controlled with charge-biased hydrogel dressings. *J Phys Chem B* 7:7184–7194
80. Venault A, Liou CS, Yeh LC, Jhong JF, Huang J, Chang Y (2017) Turning expanded poly(tetrafluoroethylene) membranes into potential skin wound dressings by grafting a bioinert epoxylated PEGMA copolymer. *ACS Biomater Sci Eng* 3:3338–3350
81. Chen YW, Venault A, Jhong JF, Ho HT, Liu CC (2017) Developing blood leukocytes depletion membranes from the design of bioinert PEGylated hydrogel interfaces with surface charge control. *J Membr Sci* 537:209–219
82. Lien CC, Chen PJ, Venault A, Tang SH, Fu Y, Dizon GV, Aimar P, Chang Y (2019) A zwitterionic interpenetrating network for improving the blood compatibility of polypropylene membranes applied to leukodepletion. *J Membr Sci* 584:148–160
83. Yeh CC, Nakagawa K, Chang Y, Hu CC (2020) Influence of photocatalysis on blood cell attachment over protein-immobilized polystyrene surfaces modified with a Poly(styrene)-b-Poly(acrylic acid) copolymer. *Langmuir* 36:3268–3275
84. Chou YN, Venault A, Wang YH, Chinnathambi A, Higuchi A, Chang Y (2018) Surface zwitterionization on versatile hydrophobic interfaces via a combined copolymerization/self-assembling process. *J Phys Chem B* 6:4909–4919
85. Venault A, Ye CC, Lin YC, Ruaan RC, Higuchi A, Chinnathambi A, Ho HT, Chang Y (2016) Zwitterionic fibrous polypropylene assembled with amphiphatic carboxybetaine copolymers for hemocompatible blood filtration. *Acta Biomater* 40:130–141
86. Venault A, Chin YT, Maggay IVB, Yeh CC, Chang Y (2022) Poly(vinylidene fluoride)/poly(styrene-co-acrylic acid) nanofibers as potential materials for blood separation. *J Membr Sci* 641:119881
87. Chen SH, Chang Y, Lee KR, Wei TC, Higuchi A, Ho FM (2012) Hemocompatible control of sulfobetaine-grafted polypropylene fibrous membranes in human whole blood via plasma-induced surface zwitterionization. *Langmuir* 28:17733–17742
88. Higuchi A, Yang ST, Li PT, Ruaan RC, Chen WY, Chang Y, Chang Y, Tsai EM, Chen YH, Wang HC, Hsu ST, Ling QD (2009) Permeation of blood cells from umbilical cord blood through surface-modified polyurethane foaming membranes. *J Membr Sci* 339:184–188
89. Yeh CC, Venault A, Yeh LC, Chinnathambi A, Alharbi SA, Higuchi A, Chang Y (2017) Universal bioinert control of polystyrene interfaces via hydrophobic-driven self-assembled surface PEGylation with a well-defined block sequence. *Macromol Chem Phys* 218:1700102
90. Sin MC, Lou PT, Cho CH, Chinnathambi A, Alharbi SA, Chang Y (2015) An intuitive thermal-induced surface zwitterionization for versatile, well-controlled haemocompatible organic and inorganic materials. *Colloids Surf, B* 127:54–64
91. Tsai CW, Hu WW, Liu CI, Ruaan RC, Tsai BC, Jin SLC, Chang Y, Chen WY (2015) The consideration of indolicidin modification to balance its hemocompatibility and delivery efficiency. *Int J Pharm* 494:498–505
92. Liao SH, Liu CH, Bastakoti BP, Suzuki N, Chang Y, Yamauchi Y, Lin FH, Wu KCW (2015) Functionalized magnetic iron oxide/alginic acid core-shell nanoparticles for targeting hyperthermia. *Int J Nanomed* 10:3315–3328
93. Tayo LL, Venault A, Constantino VGR, Caparanga AR, Chinnathambi A, Alharbi SA, Zheng J, Chang Y (2015) Design of hemocompatible poly(DMAEMA-co-PEGMA) hydrogels for controlled release of insulin. *J Appl Polym Sci* 132:42365
94. Lo CW, Liao WH, Wu CH, Lee JL, Sun MK, Yang HS, Tsai WB, Chen WS (2015) Synergistic effect of PEI and PDMAEMA on transgene expression in vitro. *Langmuir* 31:6130–6136
95. Liao WH, Hsiao MY, Lo CW, Yang HS, Sun MK, Lin FH, Chang Y, Chen WS (2017) Intracellular triggered release of DNA-quaternary ammonium polyplex by ultrasound. *Ultrason Sonochem* 36:70–77
96. Shih YJ, Venault A, Tayo LL, Chen SH, Higuchi A, Deratani A, Chinnathambi A, Alharbi SA, Quemener D, Chang Y (2017) A zwitterionic-shielded carrier with pH-modulated reversible self-assembly for gene transfection. *Langmuir* 33:1914–1926
97. Venault A, Huang YC, Lo JW, Chou CJ, Chinnathambi A, Higuchi A, Chen WS, Chen WY, Chang Y (2017) Tunable PEGylation of branch-type PEI/DNA polyplexes with a compromise of low

- cytotoxicity and high transgene expression: In vitro and in vivo gene delivery. *J Phys Chem B* 5:4732–4744
98. Chen H, Yang JT, Xiao SW, Hu RD, Bhaway SM, Vogt BD, Zhang MZ, Chen Q, Ma J, Chang Y, Li LY, Zheng J (2016) Salt-responsive polyzwitterionic materials for surface regeneration between switchable fouling and antifouling properties. *Acta Biomater* 40:62–69
 99. Zhang D, Ren BP, Zhang YX, Liu YL, Chen H, Xiao SW, Chang Y, Yang JT, Zheng J (2020) Micro- and macroscopically structured zwitterionic polymers with ultralow fouling property. *J Colloid Interface Sci* 578:242–253
 100. Venault A, Huang CW, Zheng J, Chinnathambi A, Alharbi SA, Chang Y, Chang Y (2016) Hemocompatible biomaterials of zwitterionic sulfobetaine hydrogels regulated with pH-responsive DMAEMA random sequences. *Int J Polym Mater Polym Biomater* 65:65–74
 101. Campoccia D, Montanaro L, Arciola CR (2013) A review of the biomaterials technologies for infection-resistant surfaces. *Biomaterials* 34:8533–8554
 102. Liu C, Gu XY, Cui MY, Xu Q, Li RH (2014) A novel ternary copolymerized polyzwitterionic of poly (AM/DMAM/MAE-DAPS): Synthesis and solution properties. *J Polym Res* 21:620
 103. Jhiang JS, Wu TH, Chou CJ, Chang Y, Huang CJ (2019) Gel-like ionic complexes for antimicrobial, hemostatic and adhesive properties. *J Phys Chem B* 7:2878–2887
 104. Wu RL, Xu SM, Huang XJ, Cao LQ, Feng S, Wang JD (2006) Swelling behaviors of a new Zwitterionic N-carboxymethyl-N, N-dimethyl-N-allylammonium/acrylic acid hydrogel. *J Polym Res* 13:33–37
 105. Kiesewetter MK, Shin EJ, Hedrick JL, Waymouth RM (2010) Organocatalysis: Opportunities and challenges for polymer synthesis. *Macromolecules* 43:2093–2107
 106. Cheng G, Zhang Z, Chen SF, Bryers JD, Jiang SY (2007) Inhibition of bacterial adhesion and biofilm formation on zwitterionic surfaces. *Biomaterials* 28:4192–4199
 107. Du JZ, Tang YQ, Lewis AL, Armes SP (2005) pH-sensitive vesicles based on a biocompatible zwitterionic diblock copolymer. *J Am Chem Soc* 127:17982–17983
 108. Schlenoff JB (2014) Zwitterion: Coating surfaces with zwitterionic functionality to reduce nonspecific adsorption. *Langmuir* 30:9625–9636
 109. Grigoras AG, Racovita S, Vasiliu S, Nistor MT, Dunca S, Barboiu V, Grigoras VC (2012) Dilute solution properties of some polycarboxybetaines with antibacterial activity. *J Polym Res* 19:8
 110. Keefe AJ, Jiang SY (2012) Poly(zwitterionic)protein conjugates offer increased stability without sacrificing binding affinity or bioactivity. *Nat Chem* 4:60–64
 111. Lu CC, Liu N, Gu X, Li BQ, Wang YN, Gao H, Ma JB, Wu GL (2014) Synthesis and characterization of biocompatible zwitterionic sulfobetaine polypeptides and their resistance to protein adsorption. *J Polym Res* 21:578
 112. Hiranphinyophat S, Iwasaki Y (2021) Controlled biointerfaces with biomimetic phosphorus-containing polymers. *Sci Technol Adv Mater* 22:301–316
 113. Che YJ, Tan YB, Cao J, Xu GY (2010) A study of aggregation behavior of a sulfobetaine copolymer in dilute solution. *J Polym Res* 17:557–566
 114. Kang JS, Kim J, Choi K, Hong PH, Park HJ, Kim K, Kim YK, Moon G, Jeon H, Lee SY (2021) A water-triggered highly self-healable elastomer with enhanced mechanical properties achieved using localized zwitterionic assemblies. *Chem Eng J* 420:127636
 115. Rong GL, Zhou D, Pang JH (2018) Preparation of high-performance antifouling polyphenylsulfone ultrafiltration membrane by the addition of sulfonated polyaniline. *J Polym Res* 25:66
 116. Rahman M, Brazel CS (2004) The plasticizer market: an assessment of traditional plasticizers and research trends to meet new challenges. *Prog Polym Sci* 29:1223–1248
 117. Tickner JA, Schettler T, Guidotti T, McCally M, Rossi M (2001) Health risks posed by use of di-2-ethylhexyl phthalate (DEHP) in PVC medical devices: a critical review. *Am J Ind Med* 39:100–111
 118. Zhang Y, Zhu PC, Edgren D (2010) Crosslinking reaction of poly(vinyl alcohol) with glyoxal. *J Polym Res* 17:725–730
 119. Sundaram HS, Han X, Nowinski AK, Ella-Menye JR, Wimbish C, Marek P, Senecal K, Jiang SY (2014) One-step dip coating of zwitterionic sulfobetaine polymers on hydrophobic and hydrophilic surfaces. *ACS Appl Mater Interfaces* 6:6664–6671
 120. Fang LF, Jeon S, Kakiyama Y, Kakehi J, Zhu BK, Matsuyama H, Zhao SF (2017) Improved antifouling properties of polyvinyl chloride blend membranes by novel phosphate based-zwitterionic polymer additive. *J Membr Sci* 528:326–335
 121. Lin XJ, Boit MO, Wu K, Jain P, Liu EJ, Hsieh YF, Zhou Q, Li BW, Hung HC, Jiang SY (2020) Zwitterionic carboxybetaine polymers extend the shelf-life of human platelets. *Acta Biomater* 109:51–60
 122. Chen J, Liu ZS, Nie XA, Zhou YH, Jiang JC (2018) Plasticizers derived from cardanol: Synthesis and plasticization properties for polyvinyl chloride(PVC). *J Polym Res* 25:128
 123. Kyomoto M, Moro T, Saiga K, Hashimoto M, Ito H, Kawaguchi H, Takatori Y, Ishihara K (2012) Biomimetic hydration lubrication with various polyelectrolyte layers on cross-linked polyethylene orthopedic bearing materials. *Biomaterials* 33:4451–4459
 124. Jamil MS, Ahmad I, Abdullah I (2006) Effects of rice husk filler on the mechanical and thermal properties of liquid natural rubber compatibilized high-density polyethylene/natural rubber blends. *J Polym Res* 13:315–321
 125. Zhang W, Chu PK, Ji JH, Zhang YH, Fu RKY, Yan Q (2006) Antibacterial properties of plasma-modified and triclosan or bromopol coated polyethylene. *Polymer* 47:931–936
 126. Rojas-Lema S, Lascano D, Ivorra-Martinez J, Gomez-Caturla J, Balart R, Garcia-Garcia D (2021) Manufacturing and characterization of high-density polyethylene composites with active fillers from persimmon peel flour with improved antioxidant activity and hydrophobicity. *Macromol Mater Eng*. <https://doi.org/10.1002/mame.202100430>
 127. Kliewer S, Wicha SG, Broker A, Naundorf T, Catmadim T, Streit WR, Vollstedt C, Kipphardt H (2020) Contact-active antibacterial polyethylene foils via atmospheric air plasma induced polymerisation of quaternary ammonium salts. *Colloids Surf, B* 186:110679
 128. Abd El-Rahman KM, Ali SAF, Khalil AI, Kandil S (2020) Influence of poly(butylene succinate) and calcium carbonate nanoparticles on the biodegradability of high density-polyethylene nanocomposites. *J Polym Res* 27:231
 129. Wang Y, Kim JH, Choo KH, Lee YS, Lee CH (2000) Hydrophilic modification of polypropylene microfiltration membranes by ozone-induced graft polymerization. *J Membr Sci* 169:269–276
 130. Zhao YH, Wee KH, Bai R (2010) Highly hydrophilic and low-protein-fouling polypropylene membrane prepared by surface modification with sulfobetaine-based zwitterionic polymer through a combined surface polymerization method. *J Membr Sci* 362:1–2
 131. Goli KK, Rojas OJ, Genzer J (2012) Formation and antifouling properties of amphiphilic coatings on polypropylene fibers. *Biomacromol* 13:3769–3779
 132. Huang JY, Murata H, Koepsel RR, Russell AJ, Matyjaszewski K (2007) Antibacterial polypropylene via surface-initiated atom transfer radical polymerization. *Biomacromol* 8:1396–1399
 133. Malkapuram R, Kumar V, Negi YS (2009) Recent development in natural fiber reinforced polypropylene composites. *J Reinf Plast Compos* 28:1169–1189

134. Chiu HT, Hsiao YK (2006) Compatibilization of poly(ethylene terephthalate)/polypropylene blends with maleic anhydride grafted polyethylene-octene elastomer. *J Polym Res* 13:153–160
135. Harussani MM, Sapuan SM, Rashid U, Khalina A, Ilyas RA (2022) Pyrolysis of polypropylene plastic waste into carbonaceous char: Priority of plastic waste management amidst COVID-19 pandemic. *Sci Total Environ* 803:149911
136. Venault A, Trinh KM, Chang Y (2016) A zwitterionic zP(4VP-r-ODA) copolymer for providing polypropylene membranes with improved hemocompatibility. *J Membr Sci* 501:68–78
137. Qiao YS, Zhang Q, Wang Q, Lin J, Wang JS, Li Y, Wang L (2021) Synergistic anti-inflammatory coating Zipped Up on polypropylene hernia mesh. *ACS Appl Mater Interfaces* 30:35456–35468
138. Chen SH, Chang Y, Ishihara K (2017) Reduced blood cell adhesion on polypropylene substrates through a simple surface zwitterionization. *Langmuir* 33:611–621
139. Zhang XX, Jiang XF, Qin W, Zhang K, Xin Z, Zhao SC (2021) Effect of the lanthanum and cerium phenylphosphonates on the crystallization and mechanical properties of isotactic polypropylene. *J Polym Res* 28:124
140. Huang R, Liu ZQ, Yan BY, Li YQ, Li HR, Liu DM, Wang P, Cui FY, Shi WX (2020) Layer-by-layer assembly of high negatively charged polycarbonate membranes with robust antifouling property for microalgae harvesting. *J Membr Sci* 595:117488
141. Bulanda K, Oleksy M, Oliwa R, Budzik G, Przeszlowski L, Fal J, Jesionowski T (2021) Polymer composites based on polycarbonate (PC) applied to additive manufacturing using melted and extruded manufacturing (MEM) technology. *Polymers* 13:2455
142. Regehr KJ, Domenech M, Koepsel JT, Carver KC, Ellison-Zelski SJ, Murphy WL, Schuler LA, Alarid ET, Beebe DJ (2009) Biological implications of polydimethylsiloxane-based microfluidic cell culture. *Lab Chip* 9:2132–2139
143. Belanger MC, Marois Y (2001) Hemocompatibility, biocompatibility, inflammatory and in vivo studies of primary reference materials low-density polyethylene and polydimethylsiloxane: a review. *J Biomed Mater Res* 58:467–477
144. Chen WH, Chen PC, Wang SC, Yeh JT, Huang CY, Chen KN (2009) UV-curable PDMS-containing PU system for hydrophobic textile surface treatment. *J Polym Res* 16:601–610
145. Goda T, Konno T, Takai M, Moro T, Ishihara K (2006) Biomimetic phosphorylcholine polymer grafting from polydimethylsiloxane surface using photo-induced polymerization. *Biomaterials* 27:5151–5160
146. Yeh SB, Chen CS, Chen WY, Huang CJ (2014) Modification of silicone elastomer with zwitterionic silane for durable antifouling properties. *Langmuir* 30:11386–11393
147. Zhou C, Li R, Luo W, Chen Y, Zou HW, Liang M, Li Y (2015) The preparation and properties study of polydimethylsiloxane-based coatings modified by epoxy resin. *J Polym Res* 23:14
148. Nakano H, Kakinoki S, Iwasaki Y (2021) Long-lasting hydrophilic surface generated on poly(dimethyl siloxane) with photo-reactive zwitterionic polymers. *Colloids Surf, B* 205:111900
149. L'Heureux N, Paquet S, Labbe R, Germain L, Auger FA (1998) A completely biological tissue-engineered human blood vessel. *FASEB J* 12:47–56
150. Lien WF, Liaw WC, Huang PC, Chang HL, Tsai HS (2011) Preparation of glass fiber clothes reinforced polytetrafluoroethylene film composites using plasma for polytetrafluoroethylene surface modification. *J Polym Res* 18:773–780
151. Yu CL, Yang H, Wang L, Thomson JA, Turng LS, Guan GP (2021) Surface modification of polytetrafluoroethylene (PTFE) with a heparin-immobilized extracellular matrix (ECM) coating for small-diameter vascular grafts applications. *Mater Sci Eng, C Mater Biol Appl* 128:112301
152. Wang DF, Xu YY, Lin YY, Yilmaz G, Zhang J, Schmidt G, Li Q, Thomson JA, Turng LS (2020) Biologically functionalized expanded polytetrafluoroethylene blood vessel grafts. *Biomaterials* 21:3807–3816
153. Cheng BH, Inoue Y, Ishihara K (2019) Surface functionalization of polytetrafluoroethylene substrate with hybrid processes comprising plasma treatment and chemical reactions. *Colloids Surf, B* 173:77–84
154. Chen YC, Lin HC, Lee YD (2003) The effects of filler content and size on the properties of PTFE/SiO₂ composites. *J Polym Res* 10:247–258
155. Venault A, Chang Y, Hsu HH, Jhong JF, Yang HS, Wei TC, Tung KL, Higuchi A, Huang J (2013) Biofouling-resistance control of expanded poly(tetrafluoroethylene) membrane via atmospheric plasma-induced surface PEGylation. *J Membr Sci* 439:48–57

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.